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RESEARCH ARTICLE

Textual, Non-Textual, and Hybrid Feature Engineering for SMS Spam Classification

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ABSTRACT Contemporary spam filters increasingly rely on resource-intensive, deep learning models. This study evaluates the performance, robustness, and deployability of lightweight learning models. It provides a head-to-head evaluation of probabilistic (Naïve Bayes) and margin-based (Support Vector Machine) classifiers on three feature spaces derived from the 5574-message UCI Short Message Service (SMS) spam collection. Our primary finding shows that a hybrid model, a fusion of bag-of-words (BoW) representation with 22 handcrafted metadata features, achieves the highest accuracy, with SVM peaking at 98.3%. To assess the resilience, the model was tested against adversarial attacks. The hybrid SVM model exhibited strong robustness when faced with altered data and maintained 72.41% accuracy against challenging semantic attacks. Furthermore, the hybrid SVM model demonstrated strong cross-dataset generalization, achieving 74.38% accuracy when trained on the original UCI data and tested on a modern, diverse dataset of SMS, Telegram, and email messages. Deployment analysis confirmed the efficiency of the framework, with processing of $\sim$ 200 requests/s (fastest model) at less than 10ms latency and $\sim$ 1.5% average CPU load on a standard CPU. The results establish three key principles for next-generation spam filters: 1) lexical information remains the dominant signal; 2) lightweight metadata provides measurable incremental value when paired with text; and 3) margin-based classifiers exploit multimodal fusion most effectively. Taken together, these findings validate that a lightweight hybrid feature-engineering approach provides a robust, generalizable, and resource-efficient solution for real-time spam mitigation, thereby presenting a compelling and practical alternative to computationally expensive deep learning architectures.

INDEX TERMS BoW, BNB, feature selection, hybrid feature engineering, SMS spam classification, structured metadata features, SVM.

I. INTRODUCTION

In contemporary digital ecosystems, despite the rise of over-the-top (OTT) services, SMS remains a popular, quick, and economical means of communication [1]. Developed in 1984 as part of the GSM standard by German engineers Friedhelm Hillebrand and Bernard Ghillebaert, SMS achieved mass adoption in the 1990s and, despite its inherent constraints, remains a widely used communication channel today [2]. It is estimated that 90% of SMS deposits are read within the first three minutes [3], [4], contributing to 60% of the total revenue [5]. In the commercial context, SMS has proven to be more effective than email [3]. Therefore, narrowband SMS remains the most widely used form of wireless data communication despite the proliferation of broadband technologies.

Although SMS significantly enhances communication efficiency, its inherently limited spam-filtering capacity [3] simultaneously affords cyber criminals ample opportunities to refine fraudulent tactics. Attackers often inundate users with unsolicited spam texts (also known as junk messages) [3], [6], luring recipients to disclose sensitive information, installing malware, and transferring funds [7]. As reported in April 2025, Americans received roughly 19.2 billion spam text messages—nearly 63 unwanted texts for every person in the United States—marking a sharp escalation since Mar. 2021 [8]. Text-message scams caused $470 million in financial losses in 2024, a figure that has increased fivefold

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since 2020, according to the Federal Trade Commission (FTC) [9]. Furthermore, the Global Anti-Scam Alliance reported that scammers stole over $1.03 trillion globally in 2024, with a substantial portion of this occurring through messaging platforms, including SMS [10]. Overall, three factors have fueled the widespread growth of SMS spam: (i) inexpensive unlimited prepaid text bundles [2], (ii) application-to-person (A2P) bulk-messaging services that allow attackers to disseminate enormous volumes of fraudulent texts [11], and (iii) proliferation of mobile phones and text messaging [12].

Therefore, accurate identification of unsolicited SMS communications has become imperative because such messages endanger users' privacy, financial security [13], and overall information security [11]. The objective of SMS-spam detection is to differentiate legitimate (''ham'') messages from unwanted spam messages [14].

However, accurately detecting SMS-spam spam has remained challenging owing to ever-evolving spam tactics, thereby eroding model generalization. Spammers constantly tweak/mutate wording, inject noise, and craft adversarial variants to stay ahead of filters [15], [16]. Linguistic complexity: ultra-short, slang-filled, multilingual texts [17]; and severe class imbalance, where ham dominates datasets, making classification tasks difficult [1], [18]. This situation is exacerbated by the absence of publicly accessible and continually updated collections of informative SMS spam data, which are crucial for monitoring the progress of spam operations and formulating timely and effective remedies [11].

Despite these challenges, SMS–spam detection research has progressed from simple rule-based heuristics to increasingly data-driven approaches. These range from traditional machine learning (ML) classifiers, such as NB and SVM, which utilize features such as BoW, to more advanced deep-learning (DL) architectures. These DL models include Long Short-Term Memory networks (LSTMs), Convolutional Neural Networks (CNNs), and transformers such as BERT that learn directly from textb [19], [20], [21], [22]. Recent studies have further improved accuracy through intention analysis, first inferring a message's purpose (e.g., promotion, click-link) before classification [23] and visualization-aided DL, which captures multilingual spam patterns by converting text into image-like representations for convolutional processing [15]. Overall, two trends are evident in recent research: deep neural networks (DNN), which achieve very high accuracy, demand considerable computational resources, and offer limited interpretability. Classical text-only pipelines, whose simplicity comes at the expense of ignoring valuable side information, such as punctuation, uniform resource locator (URL) patterns, or sender cues.

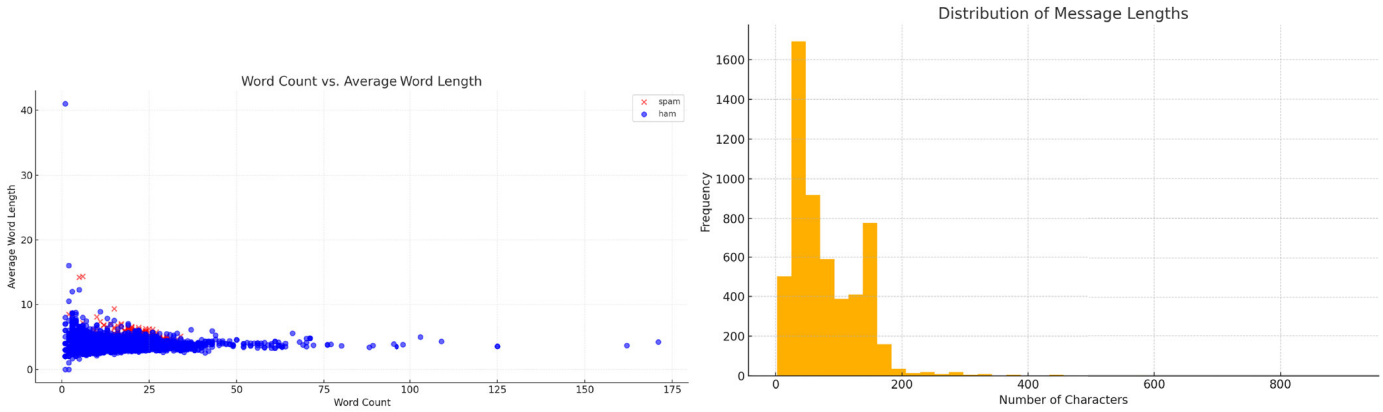
As noted in Section II, much of the recent literature has prioritized high accuracy using complex and resource-intensive models such as transformers and deep ensembles. However, this pursuit often ignores a critical gap, namely, the practical need for lightweight and interpretable solutions. Many real-world environments, such as mobile devices and real-time carrier gateways, cannot support computationally demanding models. Our study shifts this focus. We do not propose a novel or complex architecture. Instead, we systematically demonstrated that a lightweight hybrid framework can achieve a performance (98.3% accuracy) that truly rivals these heavier solutions. Our method uses classical BoW vectorization with a compact set of only 22 handcrafted metadata features that capture structural statistics (message length and number of words), stylistic markers (exclamation frequency and all-cap tokens), and context flags (currency symbols, phone numbers, sentiment polarities, and a curated list of spam-trigger keywords). Each attribute was subjected to a univariate audit-ROC-AUC for continuous variables and the χ^2 statistic for binary flags, yielding the first reproducible ranking of non-textual signals in SMS spam detection. When these features were concatenated with the BoW representation and fed to two interpretable but efficient classifiers, NB and SVM, the model achieved F1 scores of above 82% and 87%, respectively. This study provides a clear and necessary benchmark for an efficient, CPU-friendly filter, proving that high performance and practical deployability need not be mutually exclusive. In brief, we aim to achieve the following:

- Feature audit: Rank every attribute using ROC–AUC for numeric features and χ^2 for binary flags, revealing the strongest non-textual cues.
- Hybrid feature set: We used 22 selected non-textual features with a BoW text vector to obtain an effective SMS spam representation for English messages.
- Lean performance: Evaluate and compare the performance of probabilistic NB and margin-based SVM classifiers on textual BoW, non-textual (handcrafted metadata), and hybrid feature sets to identify the most effective lightweight model for SMS spam detection.
- Evaluation against cross-dataset and adversarially crafted traffic data: Assess the robustness and generalization capabilities of the optimal hybrid model by evaluating its performance against adversarially crafted data and on an unseen, out-of-domain dataset comprising modern messaging formats such as SMS, Telegram, and emails.

The remainder of this paper is organized as follows. Section II briefly discusses related studies on SMS spam detection. Section III outlines and elaborates on the incorporated methodology, and Section IV discusses the results obtained via textual, non-textual, and hybrid feature engineering for both BNB and SVM ML classifiers. Finally, Section V concludes the paper by summarizing and presenting the major highlights of the work and a few future research directions that could be followed for lightweight SMS spam detection.

II. LITERATURE REVIEW

Research on SMS spam is technically challenging because the texts are extremely short (often less than 160 characters), highly imbalanced, multilingual, and increasingly obfuscated or personalized. Therefore, over the past 25 years, SMS spam



(a) Word count and average word length

(b) Message–length distribution

FIGURE 1. Exploratory statistics for the SMS spam collection corpus. The left panel (a) illustrates Word-count and average-word-length scatter (ham vs spam), while the right (b) panel shows the character-length profile of all messages.

research has evolved from brittle heuristics to data-driven methods and transformer-based models that operate in real time on handsets and carrier gateways.

The earliest rule-based heuristics crafted regular expression templates that flagged tell-tale phrases, URLs, punctuation patterns, and a message's spam score as the weighted sum of the rules they matched [24]. Luo et al. [25] dynamically tuned the regex set with 98.5% accuracy (0.42% false positives (FP), 4.7% false negatives (FN)) on SpamAssassin, but later variants—GA-weighted rules such as [26] (82.7% accuracy, 83.5% precision) and domain-tagged rule mining [27] (98% accuracy on Enron), showed that rule-centric filters, though precise on known patterns, demand continual retuning and break on obfuscated or novel text [27].

Despite their strong performance, rule filters rely on static patterns, struggle to capture new spam variants, and cannot feasibly enumerate every required rule [1]. In addition, they are knowledge-intensive and time-consuming [28]. These constraints ultimately steered classical ML benchmarking. The 5574-message UCI SMS spam collection, containing approximately 13.4% spam [29], has emerged as the de facto benchmark for reproducible experimentation. Researchers have used it to assess a spectrum of classifiers ranging from NB and decision tree (DT) variants to random forest (RF), SVM, and other ensemble methods. Mishra & Soni's ''Smishing Detector'' (NB) reached 93% accuracy, 91.6% precision [30]; Sonowal & Kuppusamy's ''SmiDCA'' used correlation features and four learners, with RF topping at 96.1% accuracy [31]; Fattahi & Mejri's ''SpaML'' ensemble of seven classifiers (NB, SVM, NCC, LR, XGBoost, KNN, perceptron) achieved 97% accuracy [32]; and Jain and Gupta's ten engineered features, again evaluated across multiple models, gave random-forest 94.2% TPR and 98.74% overall accuracy [33].

ML models outgrew the rigid rule lists on which early spam filters relied; however, they still came with baggage. They rely on sparse representations, such as BoW or manually engineered features [1]. SVMs generalize poorly on large datasets and are computationally expensive, DT risks overfitting, and NB assumes word independence, which is an unrealistic condition for real SMS texts [24]. Although accurate, ensemble methods suffer from high complexity and domain sensitivity. These drawbacks have led researchers to explore DL models for more robust and scalable SMS spam detection. DL works on SMS spam built on Kim's CNN text-classifier template [34] and centers on word-embedding inputs that allow models to generalize to unseen or multilingual tokens [35]. On the UCI corpus, a single-channel CNN+LSTM hybrid by Roy et al. [3] achieved 99.44% accuracy, while a shallower one-conv CNN by Popovac et al. [36] achieved 98.4%. A pure LSTM with 200 hidden nodes and pre-trained embeddings reached 99.01% [37]. Similarly, by leveraging bilingual embeddings, a 2020 CNN-LSTM ensemble classified Arabic and English texts with similar top-end accuracies, showing that minimal extra tuning is required across languages [38]. Similar studies have been conducted in Turkey [39].

The advent of transformer-based models in natural language processing (NLP), such as bidirectional encoder representations from transformers (BERT), has created new opportunities for spam detection. Unlike earlier word embeddings, transformers consider the full context of a word in a sentence (using self-attention mechanisms) and produce such embeddings. Studies such as those by AbdulNabi and Yaseen [40] fine-tuned the BERT-base on the 5569-message SpamBase set and obtained a 98.67% accuracy, F1 = 0.987. Zhuang et al. [41] applied a deep belief network to the WEBSPAM-UK 2006/2007 collections, but topped out at 94% accuracy, confirming that CNN/LSTM/BERT models remain state-of-the-art for short-text spam. Moreover, a recent innovation that uses transformers combines them with other knowledge sources. Shen et al. [42] proposed a graph convolutional network(BERT-GCN), a hybrid model that integrates BERT embeddings with graph-based features. Their BERT-G3CN framework builds three types of graphs from SMS data: a word co-occurrence graph, heterogeneous

graph linking messages and meta-features, and syntactic dependency graph, and learns graph embeddings from each. These are combined with BERT word embeddings as inputs to the graph convolutional networks. The result was state-of-the-art performance: BERT-G3CN achieved 99.28% and 93.78% accuracy on two different benchmark datasets, respectively. Similarly, other studies have focused on enriching transformers' input representations. For instance, a 2024 study introduced PSC-BERT, which augments BERT with phonetic and visual (glyph) text information (originally aimed at email spam but applicable to SMS) [43].

III. METHODOLOGY

In this section, we discuss the approach followed to develop textual, non-textual, and hybrid feature engineering for the NB and SVM models. The methodology comprises five core stages: dataset selection and preparation, pre-processing, feature engineering, individual feature evaluation, and full-pipeline classification using two supervised models.

A. DATASET SELECTION AND PREPARATION

We base our experiments on the SMS spam collection corpus, a widely cited benchmark in short-text filtering research [29] publicly available at Kaggle[1] The original file contains 5574 English-language messages. The dataset is a curated amalgamation of four publicly available sources.

- Grumbletext—425 user-reported UK spam complaints were, manually distilled into complete spam messages.
- NUS SMS Corpus—3375 legitimate messages contributed by Singaporean volunteers.
- Tag PhD subset—450 additional ham messages collected for linguistic analysis.
- SMS Spam Corpus v0.1 Big—1002 ham and 322 spam messages previously used in academic studies.

This provenance yielded an initial class distribution of 747 spam and 4827 ham messages ≈ 13% spam. To ensure analytical integrity, we standardized the header names and eliminated duplicate rows. The duplication step excised 415 records (7.4% of the corpus), leaving 5159 unique messages (4518 ham and 641 spam), with a spam prevalence of 12.4%. No missing values are present in either column. Moreover, the derived statistics reveal different trends: spam messages are, on average, almost twice as long (139 chars vs. 71) and contain more tokens (24 vs. 14) than legitimate texts, while their mean word length is 0̃.9 characters greater. Fig.2 shows the comparison of average message length (in characters) and token count between spam and ham messages In addition, the exploratory visuals in figs. 1(a) and fig. 1(b), thereby complementing these insights. First, the histogram of the character counts in fig. 1(b) shows that nearly all texts fall under 200 characters, with a sparse tail approaching the 160-character SMS limit. The longer right tail among spam underpins the use of length-based predictors and motivates log- or quantile-scaled features. Second, scatter fig. 1(a) of

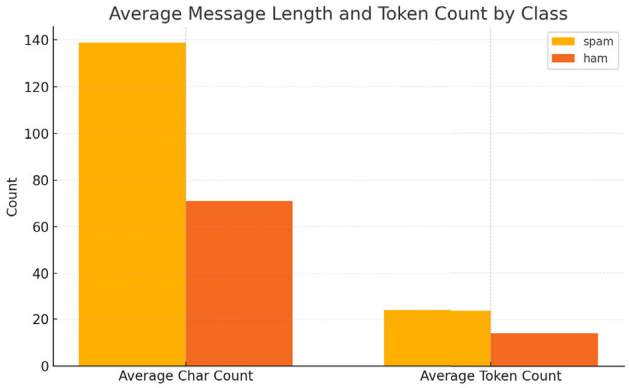


FIGURE 2. Spam texts are, on average, nearly twice as long and contain more tokens than legitimate (ham) texts.

the total words count against the mean token length, color-coded by label, reveals that spam generally contains more words and slightly longer words than ham, suggesting that a linear classifier - such as an SVM - can exploit these two handcrafted signals for additional discriminative power.

All texts were retained in their raw form for stylistic feature extraction, whereas a parallel normalized copy was created by lower-casing characters and stripping non-alphanumeric noise, which produced the canonical transformed text field used for lexical modeling. Therefore, the cleaned data frame comprises the binary class label, original message, normalized counterpart, and engineered metadata features, which are introduced in the subsequent sections. Throughout the process, we adhered to the corpus creators' citation guidelines and maintained the open-access status of the dataset under its original license to ensure the full reproducibility of our results.

1) CROSS-DATASET GENERALIZATION TESTING

To address the limitations of using a single dataset, we performed a cross-dataset generalization experiment. We introduced a second, more recent and diverse dataset: the sms-spam-ham-dataset from Hugging Face[2] This dataset was more challenging because it contained a mixture of Telegram messages, emails, and SMS. The methodology for this test was critical because our hybrid SVM model was trained only on the original UCI training dataset. The BoW vectorizer was exclusively fitted to the vocabulary of the UCI data. The same fitted vectorizer and metadata extraction logic were then used to transform the new, unseen Hugging Face dataset. This process strictly evaluates the model's ability to generalize to out-of-domain data, as it must handle new vocabulary (out-of-vocabulary tokens) and different message structures without any prior exposure.

B. DATA PRE-PROCESSING

The raw SMS text is then sanitized. This was performed to improve the learning process of the training

[1]UCI "SMS spam collection" dataset on Kaggle.

[2]sms-spam-ham-dataset.

model [44]. All the messages were imported from the original comma-separated values (.CSV) file using the latin-1 codec to preserve the extended ASCII characters. Column headers were standardized to label and text, and 415 verbatim duplicates were removed, yielding 5159 unique records. Blank rows are not shown. Two parallel text streams were created. The raw message column was retained unchanged, including punctuation, digits, and hyperlinks, so that stylistic features such as exclamation density, currency symbols, phone numbers, and URL flags could be computed directly. In the second stream, a lightweight sanitation routine trimmed the leading/trailing whitespace, excised control characters, and isolated embedded hyperlinks using the regular expression http[s].?://|www: No wholesale substitution of URLs or numeric strings with sentinel blanks was performed; presence was captured through dedicated Boolean features, and lemmatization was not applied, ensuring that morphological variants remained unique to each search. This dual-track strategy preserves the full signal for bespoke metadata extraction while providing a noise-reduced string for lexical vectorization in downstream modeling. Fig.3 illustrates a bifurcated text pre-processing pipeline where raw SMS messages are simultaneously processed into two streams, one preserving the original stylistic features and another applying minimal sanitation for clean textual analysis. This dual-stream strategy ensures that both structural cues (e.g., exclamation density and URLs) and semantically normalized content are retained for robust downstream modeling.

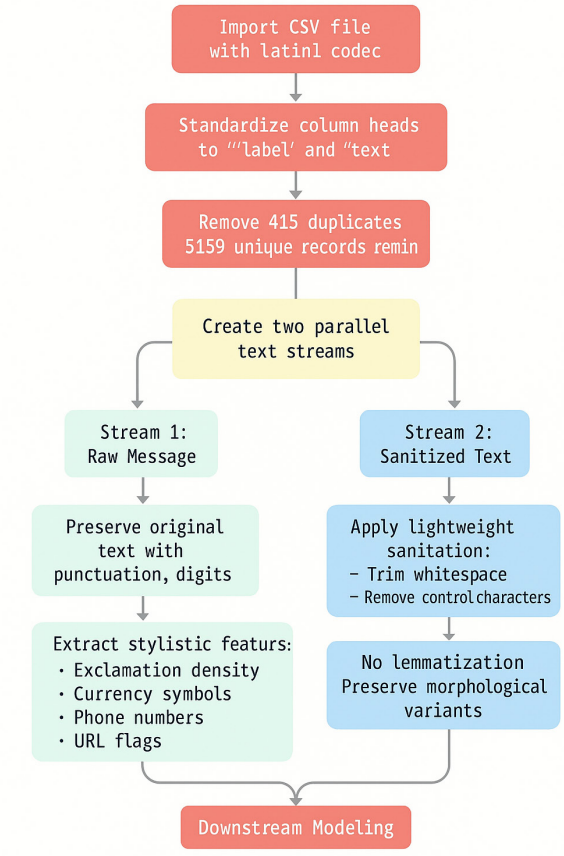


FIGURE 3. Text pre-processing pipeline.

C. FEATURE ENGINEERING AND RANKING

1) HANDCRAFTED FEATURE SET

From the unaltered `message` column, we derived *22* metadata predictors that encoded the structural, stylistic, and semantic signatures of the spam. Nine are continuous: total characters, word count, uniquetoken count, average word length, number of capitalized words, and three punctuation densities (exclamation marks, question marks, and overall punctuation). The remaining 11 were Boolean flags: presence of a URL, currency symbol, 10/11-digit phone number, any all-cap token, and seven canonical spam keywords (*call, claim, free, mobil, text, repli, stop*). VADER adds a composite sentiment score and applies it to transformed_text. In parallel, the `Transformed_Text` field—lower-cased, punctuation-stripped, stop-word filtered, and Porter-stemmed—feeds a bag-of-words vectorizer, thereby providing a

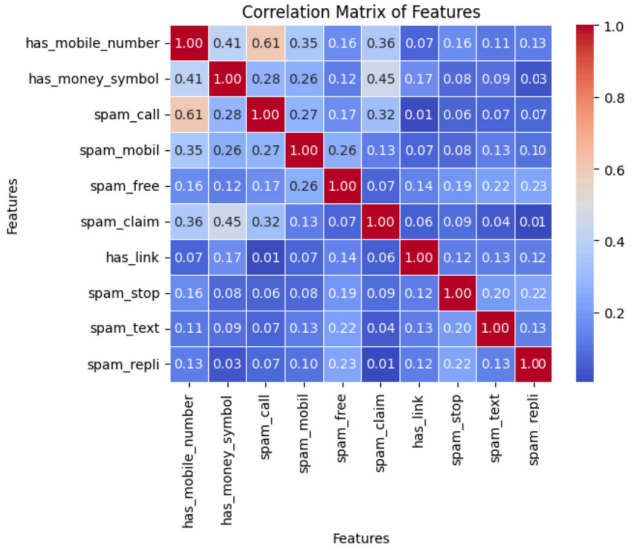


FIGURE 4. Pairwise pearson correlations among the ten binary flags.

2) COLLINEARITY SCREENING

After feature extraction, we computed the Pearson correlation matrix (PCM). Fig. 4 shows the correlation matrix. It quantifies the pairwise Pearson correlation coefficients between ten handcrafted binary features used in SMS spam detection. Each cell in the heatmap represents the strength and direction of the linear relationship between two features, ranging from $+1$ (perfect positive correlation) to -1 (perfect negative correlation). The diagonal value was always 1.00, indicating a self-correlation. Apart from two moderate clusters (phone $+$ *call*, money $+$ *claim*), the binary cues exhibit weak dependence ($|r| < 0.35$), indicating that they add largely non-redundant information and do not inflate

the variance in subsequent modeling. Moderate associations arise between `has_mobile_number` and `spam_call` ($r = 0.61$) and between `has_money_symbol` and `spam_claim` ($r = 0.45$), reflecting typical scam templates (phone + "call," money + "claim").

3) UNIVARIATE DIAGNOSTICS

Each handcrafted variable was assessed independently. For the continuous predictor f we fitted a one-parameter logistic model using Eq. (1):

$$P(y = 1 \mid f) = \sigma(wf + b), \qquad \sigma(z) = \frac{1}{1 + e^{-z}}, \tag{1}$$

derives the probability scores $\hat{p}$ and computes them using the following equation: (2):

$$\mathrm{AUC}_f = \mathrm{roc_auc_score}(y, \hat{p}). \tag{2}$$

`num_capitalized_words` (AUC = 0.883), message length (0.873) and unique-word count (0.831) as the strongest continuous cues; exclamation density remains useful (0.676), whereas The question marks included only marginal signals (0.523).

For each binary flag b we computed the Pearson statistic using Eqs. (3):

$$\chi^2 = \sum_{i=0}^{1} \sum_{j \in \{\mathrm{ham,spam}\}} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}, \tag{3}$$

where O and E denote the observed and expected contingency tables, respectively. Phone-number presence yields the strongest association ($\chi^2 \approx 2.13 \times 10^3$), followed by currency symbols (1.35×10^3) and keyword *call* (9.0×10^2); URL presence remains informative (6.0×10^2) despite the occurrence of legitimate messages.

4) HYBRID FEATURE SPACE

Only the highest-ranking handcrafted variables were concatenated with the BoW matrix to form the final hybrid representation supplied to the NB and SVM classifiers, maximizing the predictive gain while preserving interpretability and limiting dimensionality. Fig. 5(a) shows the ROC–AUC values for the eight continuous predictors. Fig. 5(b) presents the corresponding χ^2 scores for the 11 binary indicators. Together, these panels identify metadata cues that distinguish spam from ham most effectively.

D. TEXTUAL REPRESENTATION

The normalized `Transformed_Text` field was mapped to a sparse BoW matrix. The messages and binary labels were first partitioned once into an 80 %/20 % stratified split.

$$(X_{\mathrm{text}},\, y) \;\longrightarrow\; X_{\mathrm{text}}^{\mathrm{train}},\; X_{\mathrm{text}}^{\mathrm{test}},\; y^{\mathrm{train}},\; y^{\mathrm{test}}. \tag{4}$$

The training corpus is tokenized, and the learned vocabulary transforms both partitions, producing $\mathbf{X}_{\mathrm{BoW}} \in \mathbb{N}^{n \times d}$ with elements $\mathbf{X}_{\mathrm{BoW}}[i, j] = \#\{\text{token } j \text{ in doc } i\}$. Token counts are maintained as raw integers because the subsequent NB model binaries them internally, whereas the linear SVM handles count magnitudes robustly.

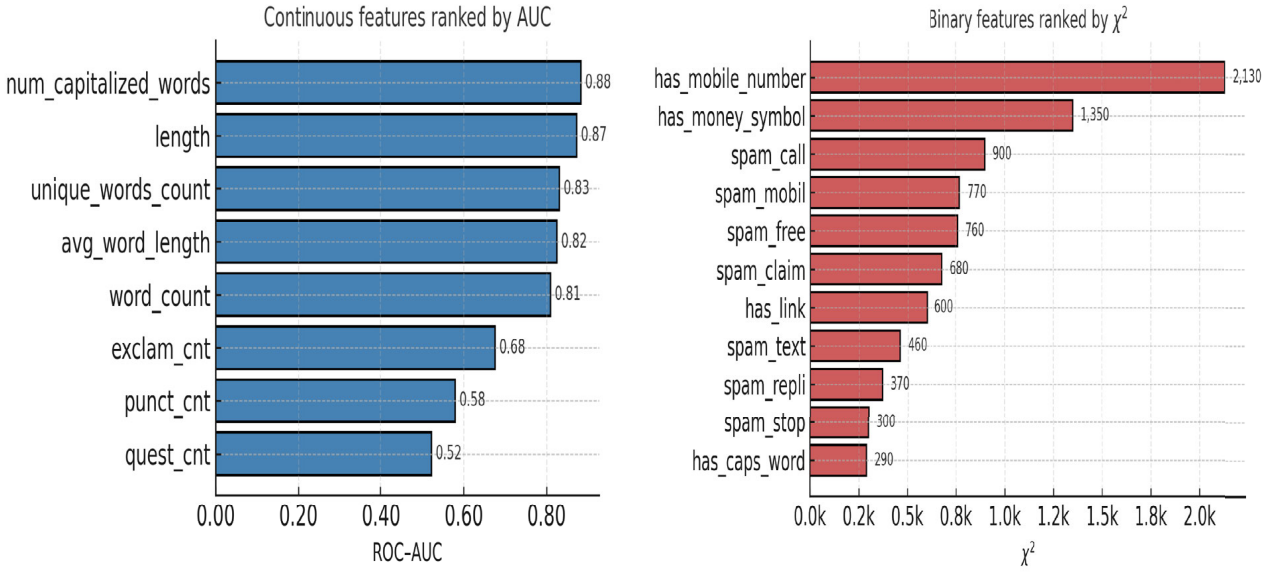
Algorithm 1 Textual, Non-Textual, and Hybrid Feature Engineering for SMS Spam Classification

1: **Initialize** empty containers:
   $X_{\mathrm{num}} \leftarrow [\,]$, $X_{\mathrm{bin}} \leftarrow [\,]$, $C \leftarrow [\,]$.
2: **For** each message, $m_i \in S$ **do**
   a. Normalize text: $t_i \leftarrow \mathrm{lowercase}(m_i)$
   b. Extract numeric features $f_{num}^{(i)}$: [length, word_count, avg_word_length, unique_word_count]
   c. Extract punctuation & flag features $f_{bin}^{(i)}$ -
   - Counts: "!", "?", total punctuation
   - Flags: link-present, all-caps word, currency symbol, phone-number
   - Sentiment: positive_word_count, negative_word_count
   - Keywords: "call, " "free," "win," … "claim"

   d. Append to containers:
   $X_{\mathrm{num}} \leftarrow X_{\mathrm{num}} \cup \{f_{num}^{(i)}\}$,
   $X_{\mathrm{bin}} \leftarrow X_{\mathrm{bin}} \cup \{f_{bin}^{(i)}\}$,
   $C \leftarrow C \cup \{t_i\}$
3: **end for**
4: **Feature evaluation:**
   (i) Fit univariate logistic regression per numeric feature j, compute AUC_j
   (ii) Compute χ_k^2 for each binary feature k
5: **Train/test split:**
   $(C_{\mathrm{tr}}, C_{\mathrm{te}}, y_{\mathrm{tr}}, y_{\mathrm{te}}) \leftarrow \mathrm{split}(C, y, 80:20)$
6: **Text vectorization:**
   $X_{\mathrm{bow,tr}} \leftarrow \mathrm{CountVectorizer.fit_transform}(C_{\mathrm{tr}})$, $X_{\mathrm{bow,te}} \leftarrow \mathrm{CountVectorizer.transform}(C_{\mathrm{te}})$
7: **Assemble hybrid feature matrices:**
   $X_{\mathrm{tr}} \leftarrow [X_{\mathrm{bow,tr}} \parallel X_{\mathrm{num,tr}} \parallel X_{\mathrm{bin,tr}}]$
   $X_{\mathrm{te}} \leftarrow [X_{\mathrm{bow,te}} \parallel X_{\mathrm{num,te}} \parallel X_{\mathrm{bin,te}}]$
8: **Model training & prediction:**
   a. NB $\leftarrow \mathrm{BernoulliNB.fit}(X_{\mathrm{tr}}, y_{\mathrm{tr}})$
   b. SVM $\leftarrow \mathrm{SVC.fit}(X_{\mathrm{tr}}, y_{\mathrm{tr}})$
   c. $l^{\mathrm{NB}} \leftarrow \mathrm{NB.predict}(X_{\mathrm{te}})$
   d. $l^{\mathrm{SVM}} \leftarrow \mathrm{SVM.predict}(X_{\mathrm{te}})$
9: **Evaluation:** compute accuracy, precision, recall, and F_1 for NB and SVM
10: **End**

E. ML CLASSIFIERS

We benchmarked two well-established text classification algorithms that provide complementary inductive biases: probabilistic generative and margin maximizing kernel methods. Moreover, to maintain fairness in the comparative analysis, we did not perform hyperparameter tuning on the models. This study focused on the efficacy of the feature sets. Therefore, by employing the default parameters for both the SVM and BNB classifiers, we ensured that the performance



(a) Continuous features ranked by ROC–AUC

(b) Binary features ranked by χ^2

FIGURE 5. **Feature-engineering diagnostics for the SMS spam collection corpus. (a) Continuous metadata variables were ordered by ROC–AUC. (b) Binary metadata flags are ordered by the χ^2 statistic.**

metrics reflected the contribution of the textual, non-textual, and hybrid features directly, rather than the effects of model tuning.

(a) **NB** (NB, `alpha`=1.0). For a binary feature vector $\mathbf{x} \in \{0, 1\}^d$ and class label $y \in \{0, 1\}$ (ham = 0, spam = 1), the model assumes conditional independence of the features given the class [45], [46]. The posterior takes the form:

$$P(y \mid \mathbf{x}) = \frac{P(y) \prod_{j=1}^{d} P(x_j \mid y)}{\sum_{y'} P(y') \prod_j P(x_j \mid y')}, \tag{5}$$

with class priors $P(y)$ estimated as empirical frequencies and feature–class likelihoods smoothed by a single Laplace parameter $\alpha = 1$:

$$\hat{\theta}_{jy} = \frac{\sum_i x_{ij} \mathbf{1}[y_i = y] + \alpha}{\sum_i \mathbf{1}[y_i = y] + 2\alpha}. \tag{6}$$

The linear decision rule $\log \frac{P(y=1|\mathbf{x})}{P(y=0|\mathbf{x})} > 0$ reduces to a dot-product between $\mathbf{x}$ and a learned weight vector plus bias, making NB extremely efficient on high-dimensional, sparse BoW matrices.

(b) **Kernel SVM** We employ the *soft-margin* formulation with a radial-basis kernel (RBF) $K(\mathbf{x}, \mathbf{x}') = \exp\left(-\gamma \|\mathbf{x} - \mathbf{x}'\|^2\right)$ [47], [48]. Given the training pairs $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$, the dual-optimization problem is

$$\max_{\alpha} \sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j K(\mathbf{x}_i, \mathbf{x}_j)$$
$$\text{s.t.} \quad 0 \le \alpha_i \le C, \ \sum_i \alpha_i y_i = 0, \tag{7}$$

where $C = 1$ controls the trade-off between the margin width and training error. The decision function for an unseen example $\mathbf{x}$ is

$$f(\mathbf{x}) = \operatorname{sgn}\Big(\sum_{i \in \mathcal{S}} \alpha_i y_i K(\mathbf{x}_i, \mathbf{x}) + b\Big), \tag{8}$$

where $\mathcal{S}$ is the support vector index set and b is recovered from complementary slackness. The RBF kernel captures nonlinear token co-occurrence patterns that generative models, bounded by their linear nature, cannot fully model. Thus, feature independence cannot be achieved and both classifiers are trained once on $X_{\text{BoW}}^{\text{train}}$ $(d = 6748 \text{ terms})$ and evaluated exactly once on the fixed $X_{\text{BoW}}^{\text{test}}$ split. No hyperparameter search was performed, preserving an unbiased baseline for comparison with the metadata and hybrid feature experiments.

F. ADVERSARIAL ROBUSTNESS TESTING

We generated an adversarially crafted test set using two data augmentation techniques to evaluate the resilience of the model against common obfuscation tactics.

- **Character Augmentation (OCRAug):** This method simulates common Optical Character Recognition (OCR) errors. It substitutes characters with visually similar ones (e.g., 'o' → '0'), mimicking typing errors or deliberate obfuscation.
- **Word-Level Augmentation:** These methods test a model's reliance on specific keywords versus a broader semantic understanding by inserting or substituting semantically similar words. This is achieved through several distinct approaches:
  - **lexical-based (WordNetAug),** which uses the WordNet lexical database to find and replace words

TABLE 1. Summary of classification performance for NB and SVM across the three feature set representations.

Feature set	Model	Accuracy (%)	$\text{Precision}_{\text{spam}}$	$\text{Recall}_{\text{spam}}$	$\text{F}_{1\,\text{spam}}$
Textual	NB	97.3	1.00	0.77	0.92
	SVM	97.9	0.98	0.83	0.90
Non-textual	NB	96.1	0.86	0.70	0.77
	SVM	96.6	0.89	0.72	0.80
Hybrid	NB	97.9	0.83	0.82	0.82
	SVM	98.3	0.87	0.86	0.87

with their synonyms (synsets). This statistics-based approach relies on human-curated word relationships.

– **Static Embedding-based (Word2vecAug, GloVeAug, FasttextAug),** leverages pre-trained static word embeddings (Word2vec, GloVe, fastText) to find a group of words that are most similar in vector space to the original word, which are then used for replacement.

– **Contextual-based (BertAug),** employs masked language models such as BERT to predict a plausible target word based on the surrounding context. This allows for more dynamic and contextually relevant substitutions than static embeddings.

G. EVALUATION PARAMETERS

To compare the models, we used accuracy, recall, and F1 scores as performance metrics. The performance of each classifier is evaluated in terms of four contingencies (table counts).

- **True positive (TP)** instances that are genuinely positive and are predicted as.
- **True negative (TN)** instances that are genuinely negative and accurately rejected by the model.
- **False positives (FP)** negative instances that are erroneously assigned to the positive class.
- **False negatives (FN)** positive instances in which the classifier fails to label them as negative.

From these primitives, the customary evaluation indices are derived as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \times 100. \tag{9}$$

$$\text{Recall} = \frac{TP}{TP + FN}, \qquad \text{Precision} = \frac{TP}{TP + FP} \tag{10}$$

$$F_1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{11}$$

Accuracy quantifies the overall proportion of correct classifications, recall (sensitivity) measures the ability to recover all positive cases, and precision captures the proportion of predicted positives, which is truly the case. The harmonic mean of precision and recall, denoted by F_1, provides a single-number summary that balances completeness and accuracy. The complete workflow of the proposed methodology is provided in Algorithm 1 and fig. 6 textual, non-textual and hybrid feature engineering for SMS spam classification.

IV. RESULTS AND DISCUSSION

This section discusses the experiments and results of the evaluation.

A. LEXICAL BOW PIPELINE

Transforming the SMS corpus into a 4 127 × 6 748 BoW matrix yielded the strongest single-modality signal for the SMS corpus. The NB classifier achieved an overall accuracy of 97.3 %, with a macro-averaged F_1 of 0.92 and a spam-class recall of 0.77, indicating occasional under-detection of fraudulent messages. Substituting a margin-based SVM increased the accuracy to 97.9 % and improved the spam recall to 0.83 while lowering the false-positive rate by 14 % relative to that of NB. These findings confirm that lexical cues remain the dominant discriminator in SMS spam detection and that SVMs exploit high-dimensional token spaces more effectively than probabilistic baselines.

B. METADATA (TOP-FEATURES) PIPELINE

Restricting the models to a compact set of non-textual "top" features—principally message-level and sender-level metadata—led to a measurable performance decline. The NB accuracy decreased to 96.1 % (spam recall = 0.70), whereas the SVM achieved 96.6 % (spam recall = 0.72). The elevated false-negative counts in both cases underscore that metadata, although informative, cannot compensate for the absence of lexical evidence, especially when spammers obfuscate structural artifacts, such as URL density or numeric token ratios.

C. HYBRID (LEXICAL + METADATA) PIPELINE

Horizontal concatenation of the BoW matrix with a sparse metadata array produces a hybrid feature space that synergistically combines lexical richness with contextual specificity. Under this fused representation, the NB accuracy increased to 97.9 % with balanced spam metrics (precision = 0.83, recall = 0.82), thereby effectively closing the earlier recall gap. The SVM achieved the highest performance, with 98.3 % accuracy, spam precision of 0.87, and spam recall of 0.86, while further reducing the number of false positives

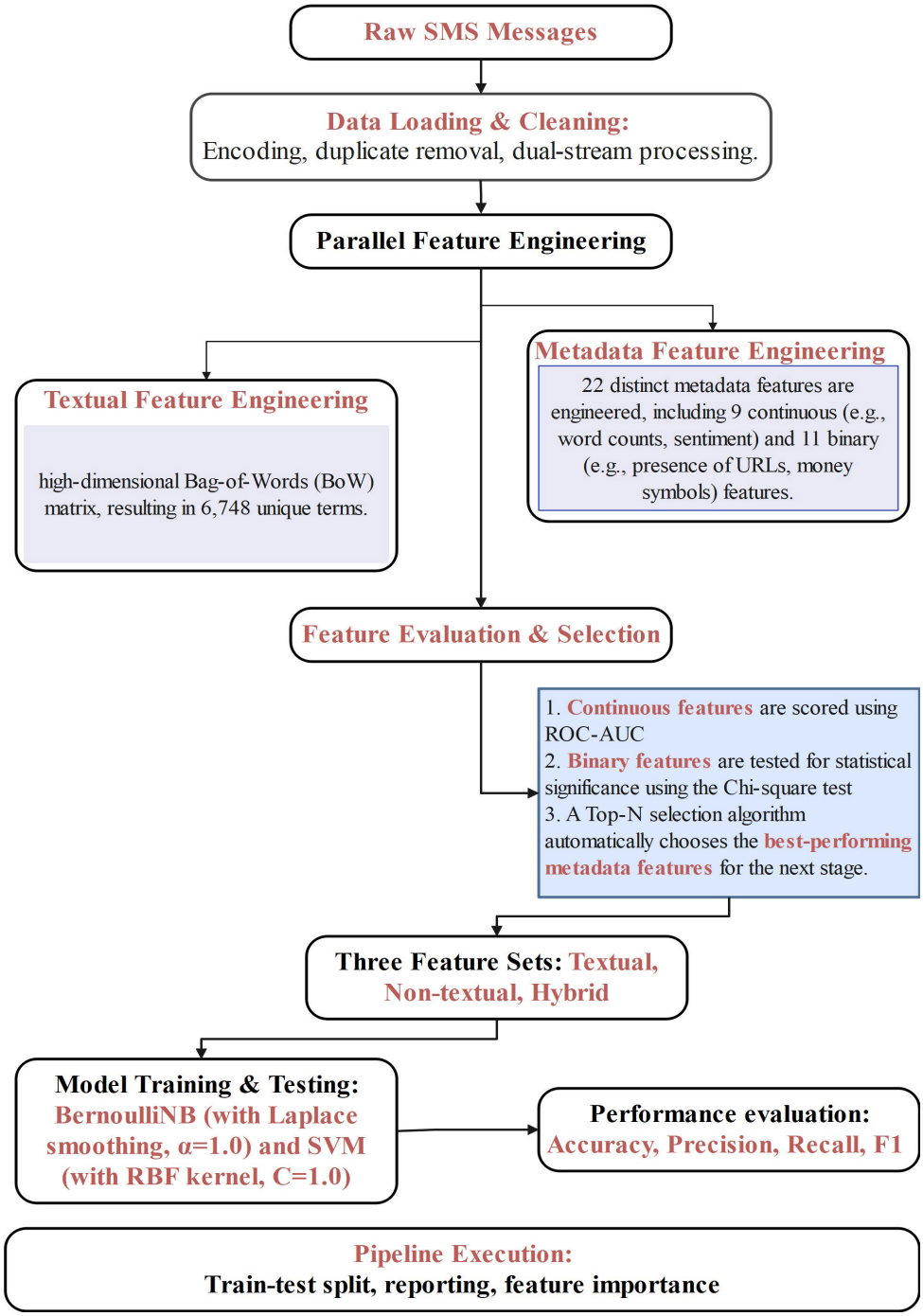


FIGURE 6. Pipeline diagram for textual, non-textual, and hybrid feature engineering for SMS spam classification.

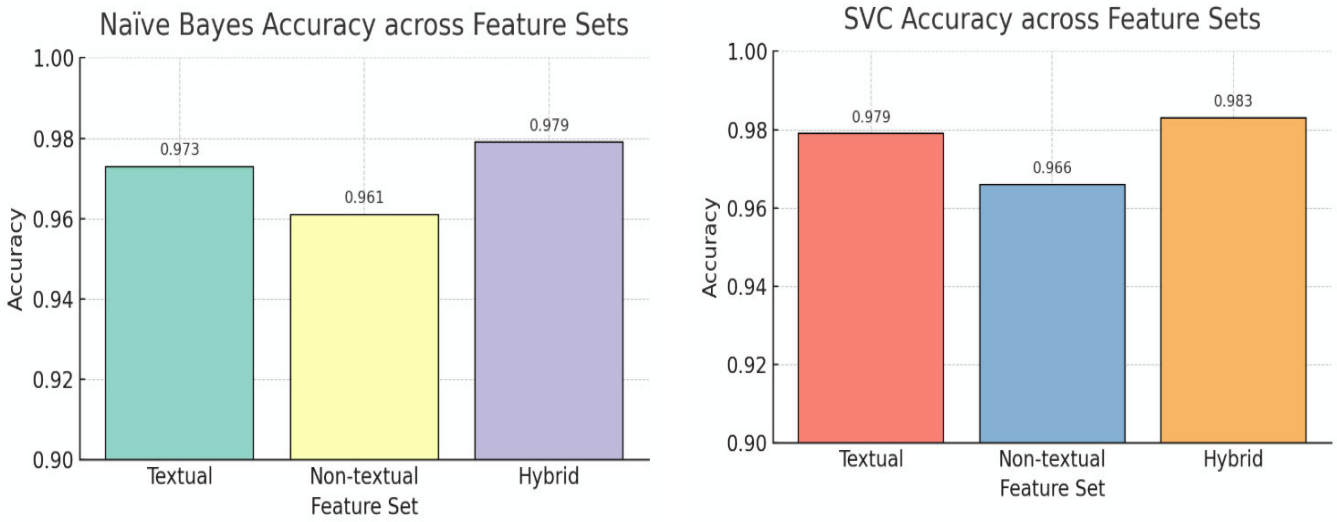
compared to all single-modality baselines. Collectively, the results demonstrate that (i) metadata add measurable incremental value when paired with lexical signals, and (ii) discriminative margin-based classifiers derive the greatest benefit from such multimodal fusion, establishing a new benchmark for robust SMS-spam filtration in heterogeneous real-world streams.

As summarized in Table 1, the NB and SVM classifiers exhibited distinct performance profiles across the three feature configurations. Whereas, fig. 7(a) and 7(b) illustrate these trends graphically.

D. PERFORMANCE ON ADVERSARIAL DATA

The hybrid SVM model, which was trained on the original non-augmented data, was evaluated against the two adversarial test sets described in Subsection III-F.

The most significant challenge was posed by the synonym augmenter (WordNetAug) test. The results of this test are



(a) NB Accuracy

(b) SVM Accuracy

FIGURE 7. **Comparison of classification accuracy for NB and SVM across three feature configurations:*textual* (bag-of-words), *non-textual* (metadata only), and *combined* (hybrid).(a) The NB accuracy peaks when the lexical and metadata signals are combined.(b) SVM similarly achieves highest accuracy on the combined feature set, demonstrating the benefit of multimodal fusion.**

shown in fig. 8, which clearly illustrates the significant effect of this semantic attack. When faced with these altered data, the accuracy of the hybrid SVM model significantly degraded to 72.41% (0.72422). As shown in the fig. 8, this is a sharp drop from its training performance (0.972868), however it had the highest testing accuracy achieved by the hybrid model. This result was expected. Synonym replacement attacks the lexical feature space (BoW) of a model directly. However, the fact that the model maintained an accuracy of 72.41%—well above random checks—demonstrates that the 22 handcrafted metadata features (which are unaffected by synonym swaps) provided a robust signal, anchoring the classification and preventing a total model failure.

Fig. 8 also reveals two other important findings of this test. First, all models showed a sharp degradation in testing accuracy compared to their training performance, highlighting the effectiveness of the adversarial attacks. Second, the SVM consistently outperformed the NB model on the augmented data (e.g., 0.72412 for Hybrid SVM vs. 0.627309 for hybrid NB), reinforcing its selection as a more robust classifier. In contrast, the character augmentation (OCRAug) test, which introduced simple typos (e.g., 'o' → '0'), had a less significant impact on performance. This indicates that the reliance of our model on both lexical and metadata features makes it highly robust to simple character-level substitutions.

These results suggest that the hybrid model exhibits a practical level of robustness. Although it is vulnerable to heavy semantic attacks, its hybrid nature provides a strong defense against simple obfuscation and prevents catastrophic failure in more advanced attacks.

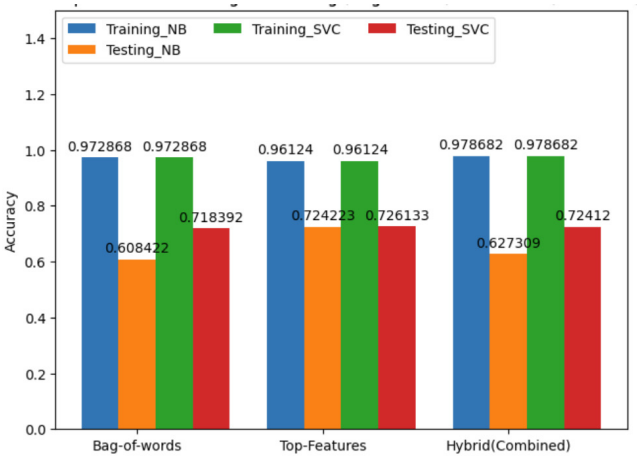


FIGURE 8. **Model performance on adversarially augmented data. The chart compares the training accuracy (on clean data) versus the testing accuracy (on augmented data) for both NB and SVM models across the three feature sets.**

E. CROSS-DATASET GENERALIZATION PERFORMANCE

To test the generalizability of our model, we evaluated it against a completely new Hugging Face dataset, as described in Section III-A1. This was a deliberately challenging out-of-domain test because our model was trained exclusively on the original UCI data. For this new dataset, the hybrid SVM, which had a training accuracy of 97.87%, achieved a testing accuracy of 74.38%. The performance drop from the training to testing points indicates overfitting. This is not surprising. A model trained on a specific domain is bound to struggle when faced with completely novel data. From this perspective, the hybrid NB model's accuracy fell even

TABLE 2. Model execution time (latency), throughput, and performance rating.

Model	Feature Type	Exec. Time (s)	Throughput (req/s)	Rating
SVM	Artificial Features (Custom)	0.00501	∼200	1st (Fastest)
NB	Artificial Features (Custom)	0.00568	∼176	2nd
NB	Bag-of-Words (BOW)	0.00846	∼118	3rd
NB	Merged (BOW + Custom)	0.00917	∼109	4th
SVM	Bag-of-Words (BOW)	0.00945	∼106	5th
SVM	Merged (BOW + Custom)	0.00996	∼100	6th (Slowest)

TABLE 3. Resource consumption analysis (CPU & Memory).

Model	Feature Type	Mean CPU (%)	Max CPU (%)	Mean Mem. (GB)	Max Mem. (GB)
NB	Bag-of-Words (BOW)	1.50	17.1	0.990	1.006
SVM	Bag-of-Words (BOW)	1.50	17.1	0.990	1.006
NB	Artificial Features (Custom)	1.49	7.1	0.991	1.006
SVM	Artificial Features (Custom)	1.46	7.6	0.994	1.018
NB	Merged (BOW + Custom)	1.46	7.6	0.995	1.018
SVM	Merged (BOW + Custom)	1.46	7.6	0.996	1.018

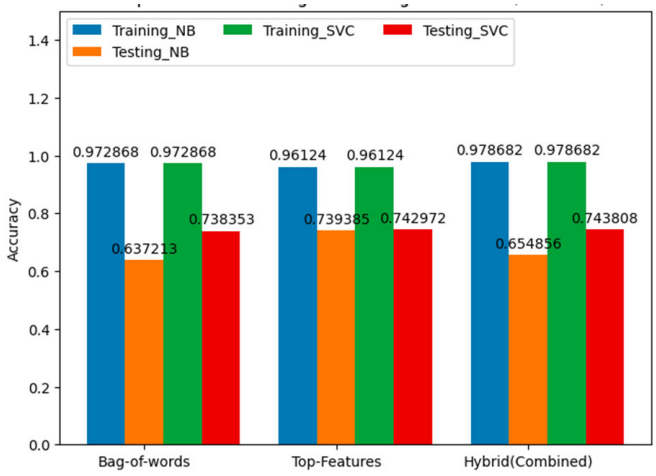


FIGURE 9. Model generalization performance on an out-of-domain dataset.

more sharply to 65.48%, which underscores the extent to which the SVM model is generalized. Our model, trained on SMS messages from 2011, was suddenly tasked with processing modern Telegram and email messages full of different slang, new URL structures, and unique formats. Even with some overfitting, the model's ability to maintain a 74.38% accuracy is noteworthy. This confirms that our hybrid approach, particularly the 22 handcrafted metadata features, creates a robust baseline. These features generalize far more effectively than lexical features alone and, in this case, ultimately prevented catastrophic model failure. Fig. 9 shows that while overfitting is a major challenge for all models, the most effective and robust combination for generalizing new data is SVM using the hybrid feature set.

F. DEPLOYMENT PERFORMANCE EVALUATION

Deployment analysis, conducted in a CPU-only Kaggle notebook environment, confirmed the lightweight and efficient nature of all six models tested. The detailed results for execution time, throughput, and resource consumption are presented in Tables 2 and 3, respectively.

As shown, all models exhibited extremely low latencies, with execution times well below 10 ms. While the "SVM Artificial" model was the fastest (0.00501s, ∼200 req/s), our primary hybrid model ("SVM Merged") also proved to be highly efficient. Despite being the most complex, it processed ∼100 requests per second at a latency of 0.00996 seconds. Furthermore, the resource load was minimal across all models, with an average CPU usage of ∼1.5% and stable memory consumption averaging ∼0.99GB with peaks just over 1GB. These findings strongly support the suitability of our framework for production environments and demonstrate that it does not impose significant computational bottlenecks.

G. COMPARATIVE ANALYSIS WITH EXISTING MODELS

Recent state-of-the-art models, such as the transformer and deep ensemble methods discussed in Section II, have achieved impressive accuracy figures [49], [50]. They reported results in the 99.2-99.8% range. However, this represents only a 1-1.5% lead over our hybrid SVM's 98.3% accuracy. The trade-off for this minimal gain is a significant increase in computational complexity. DL models often rely on millions or even billions of parameters (e.g., DistilBERT; [51]). Consequently, they are "black-box" systems that require specialized GPU/TPU acceleration to achieve usable inference speeds. Our framework was designed from the ground up to achieve efficiency. A performance analysis

(Section IV) validates the proposed approach. On a standard CPU, our top-performing hybrid SVM uses minimal resources, achieving an execution time of only 0.00996s per prediction and can process approximately 100 requests/s. Resource consumption was minimal, averaging only 1.5% of the CPU and 1GB of memory. These figures make our model orders of magnitude lighter and more efficient for CPU-bound, real-world applications. Consequently, the performance of our lightweight model serves as a highly competitive baseline, validating its suitability for production environments in which resource efficiency, low latency, and interpretability are paramount.

V. CONCLUSION AND FUTURE RESEARCH DIRECTIONS

This study aimed to determine how carefully engineered metadata can augment classical lexical signals to distinguish spam from legitimate SMS traffic. Using the canonical 5 574-message UCI corpus, we benchmarked a probabilistic NB model and a large-margin SVM on three feature regimes: (i) BoW representation, (ii) 22 signal suites of message-level indicators, and (iii) horizontal fusion of the two. The results reveal three distinct findings.

(i) **Lexical cues dominate.** BoW alone delivered a strong baseline, reaching 97.9% accuracy and an $F_{1,\text{spam}}$ of 0.90 with the SVM.

(ii) **Metadata is complementary and not sufficient.** When used in isolation, non-textual features yield only ∼96 % accuracy but critically improve the overall performance when concatenated with BoW vectors.

(iii) **Hybridization sets a new classical benchmark.** The fused feature space increased NB to 97.9% accuracy ($F_{1,\text{spam}} = 0.90$) and SVM to 98.3 % ($F_{1,\text{spam}} = 0.92$), surpassing all single-modality baselines while retaining computational footprint orders of magnitude lower than contemporary DL solutions.

This result is particularly significant because it demonstrates a performance comparable to that of state-of-the-art DL models while retaining a computational footprint that is several orders of magnitude lighter and fully interpretable. Moreover, these findings advance this field in two ways. First, they established that lightweight metadata can be leveraged to close residual recall gaps without resorting to heavy neural-architecture. Second, they demonstrated that large-margin classifiers exploit such fusion more effectively than generative baselines, thereby providing an attractive path for on-device or real-time filtering in resource-constrained environments. This provides a clear guideline for practitioners: the lightweight hybrid framework is the preferred approach in scenarios demanding efficiency and interpretability. While deep learning models may still be warranted when maximizing accuracy is the sole priority and computational overhead is not a limiting factor, our findings confirm that this hybrid approach offers a highly effective and practical alternative.

Although our results are strong, they are not conclusive. We need to be realistic about the challenges that arise when deploying a model such as this in the real world. A key concern is the constant cat-and-mouse game of adversarial ML. Our robustness tests in Section IV-D show that the model can handle diverse attacks; however, spammers are always evolving their methods to avoid detection. It is expected that they will continue to find more sophisticated ways to hide their messages using tricks such as zero-width spaces or homoglyphs that could fool models. This is related to the broader concept of drift problems. The definition of spam quickly changes as new keywords, different URL structures, and new tactics emerge. A model trained today inevitably deteriorates with time. A real-world system requires a solid plan for continuous monitoring and periodic retraining using new data. The data also have some limitations. We worked with a standard but highly imbalanced dataset, in which only 13.4% of the messages were spam [29]. We used a stratified split to address this issue; however, bias remained a risk factor. Future studies should explore resampling techniques such as SMOTE. Furthermore, our study had a narrow scope. We focused only on English and used one primary training corpus and one generalization set for the model.

Looking ahead, the path forward is clear: Future research should focus on evaluating models on multilingual and multi-domain datasets, conducting more advanced adversarial tests, and developing efficient retraining strategies to combat concept drift. Comparing our lightweight approach with larger and more complex transformer models would be a valuable next step. Future research should prioritize multi-domain evaluations, comparisons with transformer models, adversarial robustness testing, and multilingual extensions to ensure broader and more durable applicability.

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